Q3. See Quit Behavior through Logit Model

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Model

Logit(P(quit=1))=$\beta_0 + \beta_1 sex + \beta_2 dex + \beta_3 lex + \beta_4 \mathrm{lex}^2$

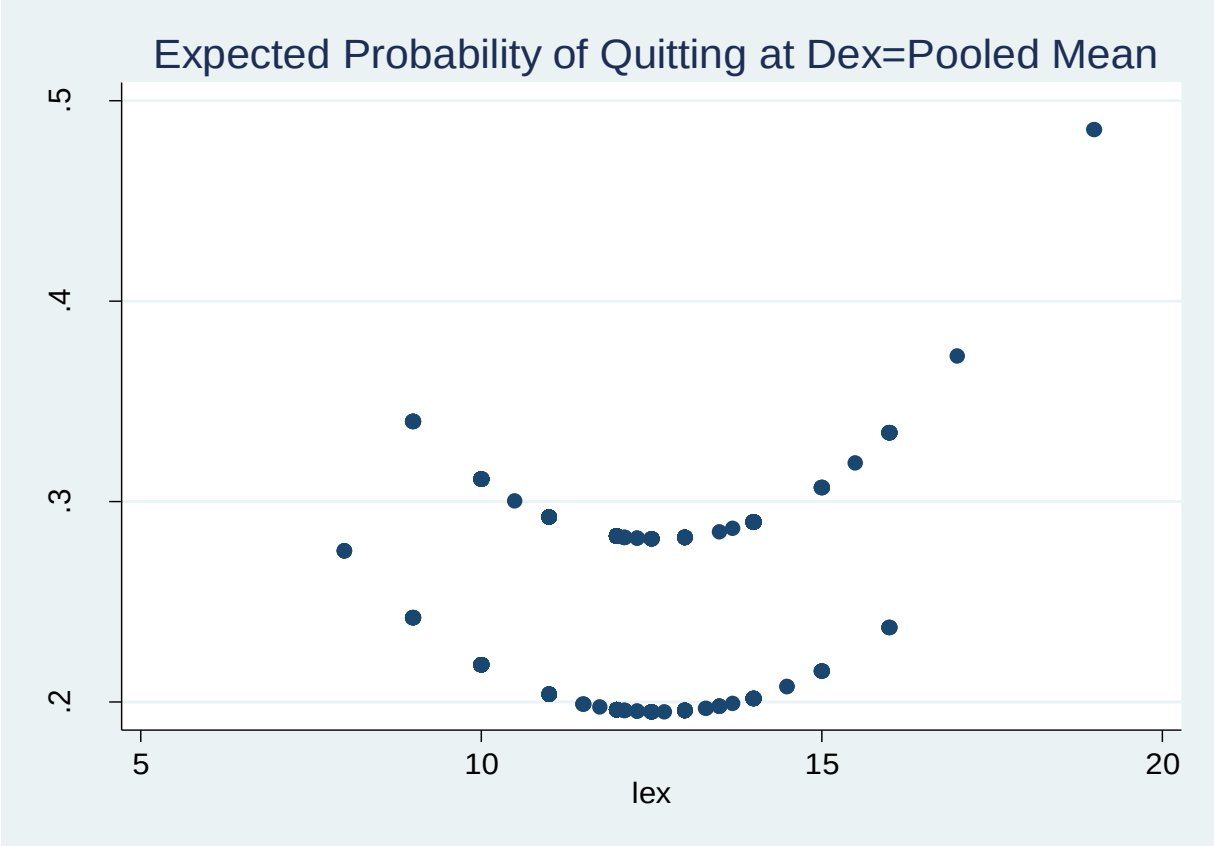
Logit regression results

	quit
sex	0.479*
	(0.1863)
dex	−0.0944***
	(0.0138)
lex	−0.539
	(0.6480)
lex2	0.0214
	(0.0259)
Constant	6.220
	(4.1047)
Observations	683

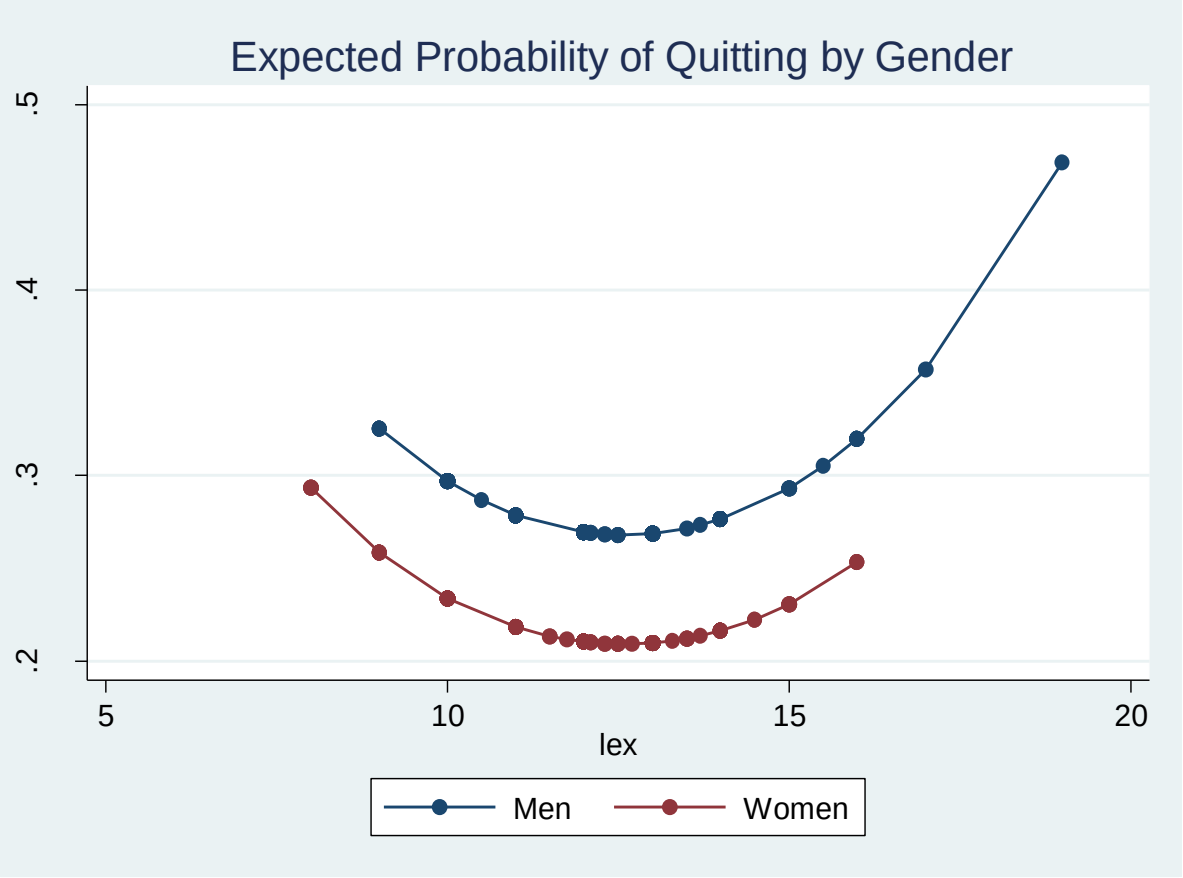
Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Pooled Mean


Expected Probability of Quitting at Dex=Pooled Mean
.5
.4
.3
.2
5
10
15
20
lex

Separate Mean



Logit Specification

- Pregibon diagnostic
- Box-Cox form of the link function is suggested as:

$$g(p) = \frac{p^{\alpha-\delta} - 1}{\alpha - \delta} - \frac{(1-p)^{\alpha+\delta} - 1}{\alpha + \delta}$$

- g(p)=log(p/(1-p)), when alpha and delta $\longrightarrow$ 0

Taylor expansion

$$g(p) = g_0(p) + \alpha g_0^{\alpha}(p) + \delta g_0^{\delta}(p)$$

$$g_0(p) = \log\left(\frac{p}{1-p}\right) = X\beta$$

$$g_0^{\alpha}(p) = 1/2[log^2(p) - log^2(1-p)]$$

$$g_0^{\delta}(p) = -1/2[log^2(p) + log^2(1-p)]$$

- Predict p
- Estimate logit belta, alpha and delta with X, $g_0^{\alpha}(\hat{p})$ *and* $g_0^{\delta}(\hat{p})$

Joint test

- H0: $\alpha = \delta = 0$ Vs.

  H1: At least one of them is not 0
- P value = 0.0015
- Reject null hypothesis
- Logit specification is not good
- Hiring and training cost